

# KAVIMUGIL RAJASEKAR

SOFTWARE ENGINEER · FLUTTER DEVELOPER · FULL-STACK · AI TOOLING

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## EDUCATION

### B.E. Computer Science & Engg.

Govt. College of Technology  
Coimbatore · 2023 – 2027  
CGPA: 7.95 / 10

## TECHNICAL SKILLS

### LANGUAGES

Python · TypeScript · Dart  
Java · JavaScript · C# · PHP  
C · Kotlin · SQL · Shell

### MOBILE & DESKTOP

Flutter 3.11+ · PyQt5  
Electron 32/42 · WPF · .NET 9  
WebRTC · nearby\_connections

### FRONTEND & BACKEND

React 19 · TanStack · Tailwind v4  
FastAPI · Express 5 · Flask  
Node.js · Prisma · Zod · Pydantic

### DATABASES & CLOUD

PostgreSQL · MongoDB · MySQL  
Firebase RTDB/FCM · SQLite  
Vercel · Cloudflare · GitHub Pages

### AI / ML

llama.cpp · Ollama · OpenRouter  
OpenCLIP ViT-B-32 · FAISS  
scikit-learn · TensorFlow · HuggingFace

### SECURITY

AES-256 EAX · PBKDF2  
HMAC-SHA256 · Linux PAM  
BlueZ RFCOMM · flutter\_secure\_storage

### EMBEDDED & IOT

ESP8266/ESP32 · MicroPython  
RPi.GPIO · MFRC522 RFID  
pyserial · ampy · esptool

### BUILD & TOOLS

Git · PyInstaller · pkg (Node)  
Inno Setup · electron-builder  
Rollup · Vitest · Playwright · uv

## LANGUAGES

Tamil (Native)  
English (Fluent)  
Hindi (Conversational)

## GITHUB

23 REPOS 14+ LANGS 20+ FRAMEWORKS

## CERTIFICATIONS

## SUMMARY

CS undergraduate at GCT Coimbatore (CGPA 7.95) with 23 public repositories spanning 8 engineering domains — mobile, desktop, AI tooling, full-stack web, CLI, IoT, and embedded systems. Builds production-ready software end-to-end: from system-level C and hardware firmware to cloud-deployed web apps and AI pipelines. Elected General Secretary, CSE Dept., GCT Coimbatore.

## EXPERIENCE

### General Secretary — Dept. of CSE, GCT Coimbatore

2023 – Present

Government College of Technology, Coimbatore

- Lead coordinator for all departmental events, inter-college competitions, and hackathons; managed logistics and student communication for 200+ student cohort.
- Primary liaison between student body and department faculty; drove academic and extracurricular initiatives for the CSE department.

## PROJECTS

### ResponseRally — Multi-LLM Benchmarking Suite

React 18 · TypeScript · Express 5 · Prisma · Neon PostgreSQL · OpenRouter

Submits one prompt to N AI models simultaneously and compares real-time latency, token throughput, and cost. JWT auth with email OTP, persistent history, custom provider registration.

### N-Queens Puzzle Studio — Flutter Social App

Flutter · Firebase RTDB/FCM · FastAPI · OpenCV · AES-CBC · Vercel Cron

Camera board scanning, MRV+forward-checking AI solver, AES-CBC QR sharing, real-time Firebase multiplayer. FastAPI backend with daily puzzle generation via Vercel Cron + FCM push.

### MySpace Connect — Offline P2P Mesh Communication App

Flutter · nearby\_connections · WebRTC · Hive · local\_auth · XOR encryption

Fully serverless peer-to-peer platform — messaging, file sharing, and WebRTC voice/video over Wi-Fi/Bluetooth mesh. Zero internet. UUID identity, XOR-encrypted QR pairing, Hive encryption.

### Splitr — Expense Settlement Engine + Flutter Wrapper

React 19 · TypeScript · TanStack Router · Recharts · Flutter WebView · file\_picker · Vercel

Debt-settlement algorithm with bidirectional cancellation, circular elimination, and greedy net-settlement (15 edge cases). Flutter Android wrapper bridges native file I/O via JS channels.

### Dupliver — AI Image Deduplication CLI

Python 3.10+ · OpenCLIP ViT-B-32 · FAISS · Typer · OpenCV · send2trash

Three-tier pipeline: SHA-256 exact, perceptual (pHash/dHash/aHash), AI-semantic (OpenCLIP + FAISS cosine). Quality-scores candidates, moves losers to OS Trash. Never auto-deletes.

### RMediTracker — RFID Healthcare Desktop System

Electron 32 · React 18 · ESP8266 · MFRC522 RFID · serialport · MongoDB Atlas

Pairs Electron desktop with ESP8266/MFRC522 firmware over USB serial. Touchless patient registration, queue management, consultation tracking. Fault-tolerant serial + MongoDB persistence.

### MunaiGuard — AES-256 File Encryption CLI

Python · AES-256 EAX · PBKDF2-HMAC-SHA256 · pycryptodomex

CLI (mg) with authenticated encryption, key derivation, rotation, SHA-256 hashing, AES benchmarking. Windows Explorer right-click integration via bundled .reg file.

### C1X-WorkFlow-Builder — Web Component npm Package

TypeScript · Custom Elements · Shadow DOM · Rollup · npm

Framework-agnostic marketing-automation workflow builder as a native Web Component. Drag-and-drop node editor, JSON import/export, undo/redo, validation. Distributed as npm package.

## ACHIEVEMENTS

- Elected General Secretary of CSE Dept., GCT Coimbatore — led all technical events for 200+ students.
- Built 23 production-quality repositories spanning 8 engineering domains within two years of study.
- Engineered cross-platform systems in C (Linux PAM module), C# (WPF/.NET 9), Kotlin (Android AlarmManager), MicroPython (ESP8266) and Raspberry Pi Flask — demonstrating deep multi-platform range.
- Explored local LLM inference (llama.cpp, Ollama), semantic embedding search (OpenCLIP + FAISS), and multi-provider AI orchestration — all applied directly in shipped, production products.
- Published C1X-WorkFlow-Builder as an npm package — a production Web Component used in marketing automation platforms, distributed via Rollup with full undo/redo and validation.

## OPEN SOURCE

- 23 public GitHub repositories — mobile (Flutter), desktop (PyQt5, Electron, WPF), AI/ML tooling, full-stack web (React + FastAPI/Express), CLI utilities, embedded (ESP8266, RPi), and security tools.
- All repositories ship with structured READMEs, architecture notes, and reproducible build steps; selected projects distributed as platform-native installers (.exe, APK, npm, standalone .exe via pkg).